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Handwritten Digit Recognition

Introduction:

Digit recognition refers to the ability of a computer to recognize and classify handwritten or printed characters like numbers, letters, symbols. Digit recognition has always been a hot topic in computer science, as it has numerous practical applications in the fields of image processing, pattern recognition, and machine learning. In this project, we will develop the basics of digit recognition, the different techniques used to recognize digits using machine learning.

How Digital Recognition Works:

Digit recognition involves preprocessing of images to enhance its quality. The image is segmented to isolate the individual characters, and then features are extracted from each character to represent its unique characteristics. These features can be defined by the number of curves, the height-to-width ratio, and the distance between certain points.

Code Report:

NOTE: Current version runs under the CPU, but can be modified to the GPU with a parameter (accelerator = "gpu") after the max_epoch argument in pytorch_lighting's Trainer.
NOTE: Deprecated functions are used, but do not dampen the demo's functionality.

This is a program that teaches the computer how to recognize digits from images. The dataset used for training & testing is the MNISIT dataset. However, there are test images that are custom drawn inside of paint.

The code organizes the transformation/model for the data inside of class and does the same thing for the data allowing efficient organization of code for testing and training. After an instance is made for these classes the computer then begins training the data calculating training loss with Cross Entropy.

After this training is done, the computer is tested with various images. Common parses of the program put the accuracy commonly over 90% with a negligible loss. After a table is displayed with the accuracy and loss an image is displayed of the kind of data in the data sets. Lastly, the images output and prediction is calculated.

Test Result 1

Test Metric	Data Loader
test_acc	0.9850999712944031
test_lost	0.04852030426263809

Table A.1 This table shows the results from the first test after epoch 5.

Conclusion:

Digit recognition is an important technology that is so useful in many fields. It involves using various techniques such as pattern recognition, machine learning to recognize handwritten or printed characters. Digit recognition is used in optical character recognition, document processing, and data entry automation, making it an essential part of modern computing nowadays.